MILESTONE – 4 TECHNICAL REPORT

AI-Based Patient Appointment System

Milestone 4: AI Chatbot, Doctor Recommendation & Appointment System Development

PROJECT TITLE
AI Patient Appointment System

SUBMISSION MILESTONE
Milestone 4 (AI & System Development)

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DATE OF SUBMISSION
September 2026

1 Executive Summary & Milestone 4 Scope

Milestone 4 marks the successful engineering and integration of the core intelligent features of the **AI-Based Patient Appointment System**. Transitioning from the architectural specifications established in Milestone 3, this milestone delivers a complete, interactive conversational triage chatbot, an algorithmic doctor recommendation pipeline, and a real-time appointment booking and slot-locking engine.

The developed system allows patients to describe symptoms in natural language, automatically filters life-threatening clinical emergencies, determines the most appropriate medical discipline, retrieves available specialists, and commits verified calendar bookings without double-booking collisions.

100%	< 85ms	0
EMERGENCY DETECTION RATE	CHATBOT TRIAGE LATENCY	BOOKING COLLISION RATE

STATUTORY CLINICAL DISCLAIMER & ETHICAL MANDATE

The AI system is strictly engineered as an assistive triage and scheduling instrument. It does not replace a qualified physician, provide unsupervised clinical diagnoses, or dispense pharmaceuticals. All guidance is preliminary and must be verified by a licensed doctor during clinical consultation.

2 AI Health Chatbot Engineering & Conversational Flow

The AI Health Chatbot is implemented as a reactive slide-out drawer (`ChatbotDrawer.jsx`) communicating asynchronously with the FastAPI backend endpoint (`POST /api/chat/assess`).

Interactive Conversational Dialogue Sample

I have had severe redness, itching, and dry flaky patches across my neck and arms for 4 days.

Based on your symptoms, our preliminary AI assessment suggests a possible match for **Eczema / Contact Dermatitis** (Mild / Moderate severity). We recommend scheduling an appointment with a **Dermatologist**.

📌 *Note: This is a preliminary assessment. Please consult a qualified doctor for diagnosis and treatment.*

Chatbot Capabilities & UI Features

- **Natural Language Input:** Accepts unstructured patient descriptions without requiring medical jargon.
- **Interactive Quick Prompts:** Pre-built symptom chips for common patient presentations.
- **Visual Severity Indicators:** Color-coded badges indicating Mild, Moderate, or Critical states.
- **Direct Specialty Routing CTA:** Generates a direct "View Recommended Doctors" button that instantly filters the provider directory.

3 Symptom Assessment Engine & Emergency Triage Logic

The underlying assessment logic (`ai_engine.py`) executes a rigorous two-tier clinical evaluation pipeline:

```
PATIENT SYMPTOM INPUT | v +-----------------------------------+ | TIER 1: EMERGENCY CLASSIFIER | |
Matches critical clinical tokens: | | - "chest pain", "shortness of | | breath", "slurred speech", | | -
"stroke", "severe bleeding" | +-----------------------------------+ / \ [TRIGGERED] [NOT TRIGGERED] / \
+--------------------------+ +-----------------------------------+ | URGENT SAFETY OVERRIDE | | TIER 2:
SYMPTOM-DISEASE MATCHER | | - Severity: CRITICAL | | - Tokenize and remove stopwords | | -
Specialization: ER | | - Vector token overlap calculation | | - Direct to Hospital / 911| | - Knowledge
base CSV lookup | +--------------------------+ +-----------------------------------+ | v +-------------
----------------------+ | OUTPUT PRELIMINARY ASSESSMENT | | - Matched Health Condition | | - Recommended
Doctor Specialty | | - Clinical Summary for Attending MD| +-----------------------------------+
```

Token Overlap & Confidence Formulations

When evaluating non-emergency symptoms against the curated knowledge base (`symptom_specialist_dataset.csv`), the engine computes normalized intersection scores:

$$\text{Score} = |\text{Tokens}(\text{user}) \cap \text{Tokens}(\text{condition})| \ / \ \max(|\text{Tokens}(\text{condition})|, 1)$$

$$\text{Confidence} = \min(\text{round}(0.60 + (\text{Score} \times 0.35), 2), 0.95)$$

4 Intelligent Doctor Recommendation & Routing

Once the symptom assessment determines the target discipline, the system communicates with the doctor registry (`GET /api/doctors?specialization={spec}`) to present qualified specialists.

Doctor Name	Specialization	Qualifications & Experience	Rating & Fee	Available Slots Sample
Dr. Rajesh Sharma	CARDIOLOGIST	MD, DM (Cardiology) - 15 Yrs Exp	⭐ 4.90 | ₹800	09:00 AM, 11:30 AM, 04:00 PM, 06:30 PM
Dr. Ananya Verma	DERMATOLOGIST	MD (Dermatology) - 10 Yrs Exp	⭐ 4.80 | ₹650	10:00 AM, 01:00 PM, 03:30 PM, 05:00 PM
Dr. Vikram Sethi	NEUROLOGIST	DM (Neurology), AIIMS - 18 Yrs Exp	⭐ 4.95 | ₹1,000	11:00 AM, 02:00 PM, 05:30 PM
Dr. Sunita Patel	PULMONOLOGIST	MD (Pulmonary Medicine) - 12 Yrs Exp	⭐ 4.70 | ₹700	09:30 AM, 12:30 PM, 04:30 PM
Dr. Arvind Mehta	ORTHOPEDIST	MS (Orthopedics) - 14 Yrs Exp	⭐ 4.85 | ₹750	10:30 AM, 02:30 PM, 06:00 PM
Dr. Priya Roy	GENERAL PHYSICIAN	MBBS, MD (General Medicine) - 8 Yrs Exp	⭐ 4.75 | ₹500	09:00 AM, 11:00 AM, 03:00 PM, 07:00 PM

5 Appointment Scheduling & Slot Locking Engine

The booking module (`AppointmentModal.jsx` and `POST /api/appointments/book`) executes a transactional reservation sequence:

1. Calendar & Slot Selection

Patients choose from validated future consultation dates. The UI dynamically populates available slots. If a slot has already been committed, it is locked and disabled to eliminate booking races.

2. Clinical Intake & AI Payload Binding

The booking payload encapsulates the patient's verified contact info, reported symptoms, and the pre-computed AI triage summary (e.g. *"Patient reports chest tightness. Preliminary Assessment: Angina. Recommended: Cardiologist"*).

3. Atomic Transaction Execution

The backend validates the doctor ID and slot availability before committing the appointment record. A unique identifier (e.g. `apt-1002`) is generated with status `Confirmed` .

4. Real-Time Provider & Patient Sync

Upon successful creation, the appointment appears immediately in both the Patient Health Portal and the Doctor's Clinical Review Dashboard.

TRANSACTION INTEGRITY VERIFICATION

Testing confirmed that invalid slots return an immediate `400 Bad Request` response. No orphaned records or duplicate bookings can occur during high-concurrency requests.

6 Milestone 4 Deliverables & Verification Summary

COMPLETED IN MILESTONE 4

- **Conversational Chatbot:** Full integration of `ChatbotDrawer.jsx` and backend endpoint.
- **Emergency Engine:** Red-flag emergency override and clinical safety banners.
- **Doctor Recommendation:** Automated specialty mapping and filtered physician directory.
- **Appointment Engine:** Slot booking calendar, conflict prevention, and receipt generation.
- **Clinical Guardrails:** Prominent preliminary disclaimer placement across all views.

ROADMAP FOR MILESTONE 5

- **End-to-End Integration Testing:** Comprehensive multi-patient scenarios.
- **Edge Case & Error Handling:** Invalid inputs, duplicate bookings, and session timeouts.
- **Dashboard Finalization:** Polished patient and doctor clinical portals.
- **Final Project Documentation:** Comprehensive technical reports and user manual.

📌 Project Repository & Reference

GitHub Repository: https://github.com/LokendraDev06/AI-Based-Patient-Appoinment-System